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GMSK Demodulation Combining 1D-CNN and Bi-LSTM Network Over Strong Solar Wind Turbulence Channel

Manqin Zhu[1], Shengbo Hu[1,2], Kai Yang[3], Tingting Yan[1], Lv Ye[1], Hongjuan Li[1], and Yang Jin[1]

[1]School of Big Data and Computer Science, Guizhou Normal University, Guiyang, China, [2]National Space Science Center, Beijing, China, [3]School of Big Data and Information Engineering, Guizhou University, Guiyang, China

Key Points:

- The K distribution channel model for the strong solar wind turbulence is innovatively established by analyzing the model of telemetry signal amplitude fluctuation caused by solar wind turbulence
- To demodulate the Gaussian minimum frequency shift keying (GMSK) signals over strong solar wind turbulence channel, two neural network demodulators based on bidirectional long short-term memory network (LSTM) and one-dimensional convolutional neural network (1D-CNN) are designed, which can obtain better bit error rate performance than the conventional coherent demodulation
- By combining the feature extraction capability of the 1D-CNN and the processing capability of LSTM for time series, we propose a novel GMSK demodulator

Correspondence to:
S. Hu,
hsb@nssc.ac.cn

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Abstract To solve the problem of bit error rate (BER) performance degradation over strong solar wind turbulence channel, this paper addresses to Gaussian minimum frequency shift keying (GMSK) demodulation using machine learning. First, by analyzing the scintillation characteristics of the telemetry signal caused by solar wind turbulence during the solar superior conjunction, the K distribution channel model is innovatively established over strong solar wind turbulence channel. Then, the approximate probability density function of the K distribution of the established channel is studied by using Laguerre orthogonal polynomial for convenience. Additionally, the BER performance of GMSK over strong solar wind turbulence channel is analyzed. Second, using a one-dimensional convolutional neural network (1D-CNN) and bidirectional long short-term memory network (Bi-LSTM), two demodulators for GMSK are proposed. By integrating the 1D-CNN and LSTM to extract local features and process the dynamic information of GMSK signals, we propose a novel GMSK demodulator over strong solar wind turbulence channel with combining 1D-CNN and Bi-LSTM. The simulation results show that, even if the solar wind turbulence has serious effects on the signal at a low signal-to-noise ratio, the proposed combining neural network demodulation can obtain better BER performance than the classical Viterbi algorithm, demodulation using only a single 1D-CNN or Bi-LSTM. The proposed GMSK demodulator is more suitable for the deep space exploration of the solar system.

Plain Language Summary The solar wind turbulence is formed by many irregular plasmas ejected from the solar nuclear reaction, and it has a serious impact on the deep-space communication link. To solve the problem of bit error rate (BER) performance degradation by caused solar wind turbulence, we analyze the scintillation characteristics of the telemetry signal caused by solar wind turbulence during solar superior conjunction. The K distribution channel model is established for strong solar wind turbulence channel. Also, for the Gaussian minimum frequency shift keying (GMSK) modulator by the Consultative Committee for Space Data Systems, we present the BER performance of the GMSK demodulator using Viterbi algorithm, only a single one-dimensional convolutional neural network (1D-CNN) and a bidirectional long short-term memory (Bi-LSTM) neural network. For different scintillation indices, the simulation results test the validity of demodulation using machine learning. In particular, by combining the feature extraction capability of the 1D-CNN and the processing capability of LSTM for time series, we propose a novel GMSK demodulator that combines the 1D-CNN with Bi-LSTM, and the BER performance using the proposed GMSK demodulator improves better even at a low signal-to-noise ratio.

1. Introduction

During the deep space exploration of the solar system, the reliable communication link is vital, since it is the only way that enables the transmission of telecommand from the earth and the return of telemetry information and scientific data from the probe. However, the communication link in the solar system space environment faces many severe challenges. In addition, due to the ultralong communication distance, the communication link between the Earth and the solar system probe may encounter solar scintillation due to solar wind during near superior solar conjunction (Morabito & Hastrup, 2001).

Solar scintillation is a telecommand and telemetry signal fluctuation phenomenon. When the transmitted telecommand and telemetry signals pass through the solar corona region, the solar wind causes telecommand and telemetry signal amplitude fluctuation, phase fluctuation and spectrum expansion. Especially, if the solar scintillation is sufficiently strong that means the heliocentric distance is less than four solar radii(Xu & Song, 2019), the communication link may be intermittently or even completely interrupted, which results in communication failure

(Stocker et al., 2018; Xu & Song, 2019). Therefore, it is critical to analyze and design a reliable communication system for solar system exploration. In fact, as the exploration of solar system has intensified (Deutsch, 2020), research on the effects of solar wind turbulence on communication system has become a hotspot. Recently, research has mainly focus on the follows:

Analyzing the performance: Among them, Xu and Song (2016) presented a theoretical study of the signal fluctuations including an analytical expression for the amplitude fluctuations. Xu and Song (2019) analyzed the effects of solar scintillation on deep space communications by using an effective solar scintillation prediction model. However, most works focus on analyzing the bit error rate (BER) performance by using binary phase shift keying (BPSK) demodulation under the condition of weak solar scintillation, with no analyzing of other demodulation techniques (Deutsch, 2020; Morabito, 2003; Morabito & Hastrup, 2001; Pan et al., 2018; Stocker et al., 2018; Xu & Song, 2019; Yao et al., 2018).

Modeling channel: investigated and discussed a solar coronal turbulence channel model. Yao et al. designed a communication link parameter influence of solar scintillation. Although the influence of solar wind on the BER performance was analyzed in the work, it does not give a closed BER, or improve the BER performance. In addition, based on the characteristics of the environment, Pan et al. (2018) divided deep space channels into three types, near-Earth links, interstellar links and near planet links. For the interstellar, using the relationship between scintillation index and Rician factor found by Morabito in 2003, the Rician channel model is established over weak solar scintillation based on the Born approximation (Morabito 2003, 2007; Qi et al., 2012). Xu and Song (2016) used Rytov first-order approximation. The works show that solar wind physical parameters are related to turbulence channel parameters. Inspired by this, we explore strong solar wind turbulence channel.

The statistical model to describe atmospheric turbulence channel has recently become a research hotspot. In (Parry & Pusey, 1979), the K distribution model was used to describe the atmospheric turbulence channel, and the closed BER expression of the free-optical space (FSO) link was given. And Le et al. (2019) studied the error control problem of FSO communication under a gamma-gamma (ΓΓ) distribution atmospheric turbulence channel. Additionally, Peppas et al. (2012) analyzed the average capacity of an optical wireless communication system over an I-K atmospheric turbulence channel. From the consistency of theoretical and experimental data, Sandaldis et al. (2008) thought that the K distribution is more suitable to describe atmospheric turbulence channel. In fact, solar wind also has characteristic turbulence. Hence, we address to the solar wind turbulence channel modeling by using the K distribution and the BER performance analysis in this paper.

Meanwhile, machine learning (ML) has been increasingly used in communication, especially in signal demodulation. Among them, Wu (2019) designed a neural network demodulator for a various modulation signals, which can perform demodulation by loading certain parameters according to the modulation scheme and not changing the receiver hardware. Mohammad et al. (2018) proposed a binary frequency shift keying (BFSK) signal demodulator based on deep neural network, and the results show that it is superior to other ML classifiers. Zhang et al. (2018) proposed a BPSK demodulator based on one-dimensional convolutional neural network (1D-CNN). Two cascaded 1D-CNNs are used to extract features from the phase information of the modulation signal, which can obtain better BER performance than the traditional Viterbi algorithm. In addition, Elbaz et al. (2018) proposed that using the long and short memory network (LSTM) to learn mapping directly from baseband modulated signals to modulated audio at the transmitter, which would create baseband to speech mappings. This method is proven to perform better than the traditional receiving system at low signal-to-noise ratio (SNR) conditions. However, in these literature, the neural network demodulators are designed only for additive white Gaussian noise (AWGN) or Rayleigh-faded channels, and the demodulation for the K distribution channel is not considered. Therefore, we intend to explore a new demodulation technique in solar scintillation channel by combining 1D-CNN and LSTM to futher improve the BER performance. Particularly, considering the Gaussian minimum frequency shift keying (GMSK) modulation is the main method recommended by the Consultative Committee for Space Data Systems (CCSDS) (Sessler et al., 2007), we are committed to GMSK demodulation in this paper.

In summary, to the best of our knowledge, this is the first work about GMSK demodulation based on ML over strong solar wind turbulence channel. The main contributions of this paper are summarized as follows:

- Compared with previous Rician channel model under the condition of weak solar scintillation (Morabito 2003, 2007), the K distribution channel model for strong solar wind turbulence to further investigate the coronal structure is innovatively established by analyzing the model of telemetry signal amplitude fluctuation caused by solar wind turbulence. Also, using the Laguerre orthogonal polynomial, the K

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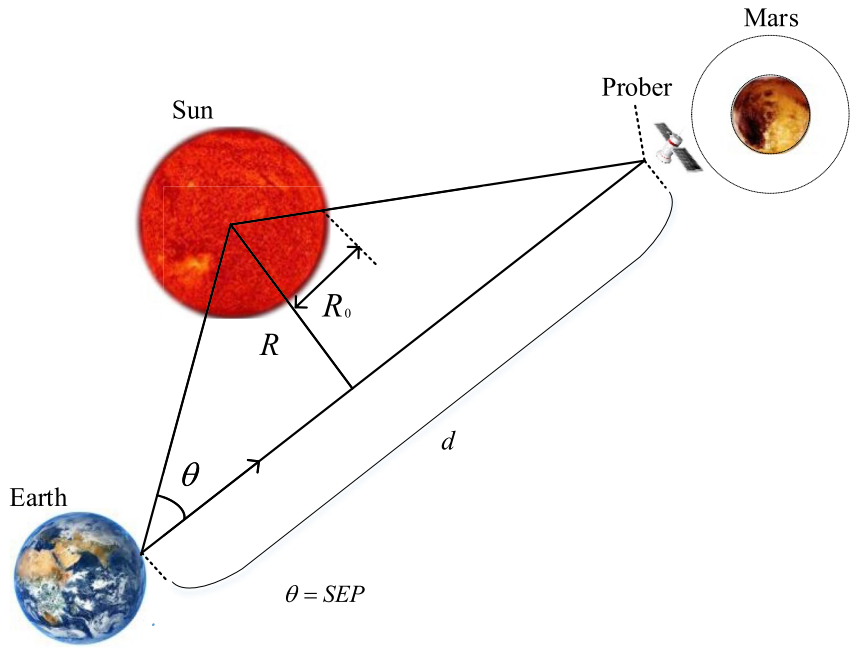

Figure 1. Superior solar conjunction geometry diagram.

distribution is approximated, and the validity of the approximate distribution is proven by Kullback-Leibler (KL) divergence.

- To demodulate the GMSK signals over strong solar wind turbulence channel, two neural network demodulators based on 1D-CNN and bidirectional long short-term memory network (Bi-LSTM) are designed, which can obtain better BER performance than the classical Viterbi algorithm.
- Especially, combining the feature extraction capability of 1D-CNN and the processing capability of LSTM for time series, we propose a novel GMSK demodulator. Simulation results show that the proposed demodulation with combining neural network can achieve better BER performance than the classical Viterbi algorithm, single 1D-CNN and Bi-LSTM over strong solar wind scintillation turbulence channel. In addition, we obtain public data set from IEEEDataPort to validate the proposed model, whose simulation results show that the proposed model still achieves better BER performance. This indicates that the neural network model has greater potential in the demodulation of wireless fading signals.

The rest of this paper is organized as follows: Section 2 introduces the model of telemetry signal amplitude fluctuation caused by solar wind turbulence, and gives the K distribution channel model. In Section 3, the BER performance of both GMSK and BPSK demodulation using the classical Viterbi algorithm are analyzed under strong solar wind turbulence. Section 4 describes the two demodulation using 1D-CNN and Bi-LSTM and proposes a novel demodulation combining 1D-CNN and Bi-LSTM. In Section 5, the BER performance using the proposed demodulators is analyzed. Section 6 concludes the paper.

2. A Channel Model Caused by Strong Solar Wind Scintillation

2.1. Geometric Model During Superior Solar Conjunction

During superior solar conjunction, the solar wind turbulence formed by many irregular plasmas ejected from the solar nuclear reaction has a serious impact on the deep-space communication link, which increases the BER, and even communication failure. The effect of solar wind turbulence on the signal is related to the Sun-Earth-Probe angle (SEP in Figure 1). The relevant geometry of the Earth-to-probe radio link relative to the Sun is described in Figure 1, where R_0 is the solar radius, R is the minimum distance from the Sun to the signal raypath, which is usually measured in the number of solar radius, and d is the total signal raypath distance.

The scintillation Index: m is used to describe the extent of amplitude scintillation. It can be defined as the ratio of the root mean square of the received signal strength fluctuation to its mean value, which is expressed as follows (Ishimaru, 1991).

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$$m = \frac{\left\langle I^2 - \langle I \rangle^2 \right\rangle}{\langle I \rangle^2} \tag{1}$$

where $\langle\rangle$ is the ensemble average, and I is the received signal strength, that is, the square of the received signal amplitude. The solar wind scintillation is weak solar scintillation when $m < 1$ and strong solar scintillation when $m \geq 1$ (Shishter 2021a, 2021b).

2.2. The K Distribution Channel Model Over Strong Solar Wind Turbulence

Deep space exploration missions require the modeling of deep space communication channels. Due to the turbulence nature of space channels, propagating electromagnetic waves suffer from fading-induced turbulence. Morabito analyzed the signal amplitude data received from deep space mission in the event of solar superior conjunction. The Rician distribution is consistent with the data for various m values. However, the Rician distribution is limited to unity m. For m up to$\sqrt{2}$, the Nakagami distribution is found to be consistent with data. Although Shishter given the probability density of irradiance for electromagnetic waves that propagate through the solar corona, the proposed model is not suitable for strong solar wind turbulence channel with all $m \geq 1$ (Shishter 2021a, 2021b). Meanwhile, the K distribution to describe strong turbulence has been successfully applied to atmospheric scattering and ionospheric scintillation (Kiasaleh, 2006; Parry & Pusey, 1979). The strong solar wind turbulence channel is described with the K distribution inspired in this paper.

For the deep space exploration communication system, the receiving signal can be expressed as:

$$r(t) = \sqrt{p}h(t)x(t) + n(t) \tag{2}$$

where p is the power of the deep space prober transmitter, $x(t) \in \{0, 1\}$ is the transmitted signal, $h(t)$ is the turbulence channel gain coefficient caused over strong solar wind turbulence, and $n(t) \sim N\left(0, \sigma^2\right)$ is AWGN, and σ^2 is the noise variance.

Inspired by the atmospheric turbulence channel, the strong solar wind turbulence channel is described with K distribution. The probability density function (PDF) is given by (Tekel & Cohen, 2012).

$$f(h) = \frac{4b^{(\alpha+1)/2}}{\Gamma(\alpha)} h^{\alpha} B_{\alpha-1}(2\sqrt{b}h) \tag{3}$$

where $B_n(\cdot)$ is the modified Bessel function of the second type of order, $b = \alpha/\mu$, μ is the intensity of the signal amplitude fluctuations, α is the solar wind turbulence channel parameter. Additionally, the strong solar turbulence with the scintillation index $m > 1$ and positive parameter α, the relationship of α with and m is given by (Samimi, 2012).

$$\alpha = \frac{2}{m-1} \tag{4}$$

2.3. The K Distribution Approximation of Strong Solar Wind Turbulence Channel

To further simplify the complex distribution and help design the wireless communication system for the deep space system of the solar system, we use Laguerre orthogonal polynomial method to approximate the K distribution.

Approximate K distribution of the strong solar wind turbulence channel:

For any positive number β, the weighting function is

$$w(\beta h) = \beta e^{-\beta h} \tag{5}$$

where

$$f(h) = \beta e^{-\beta h} \sum_{i=0}^{\infty} (L_i(\beta h)) L_i(\beta h) \tag{6}$$

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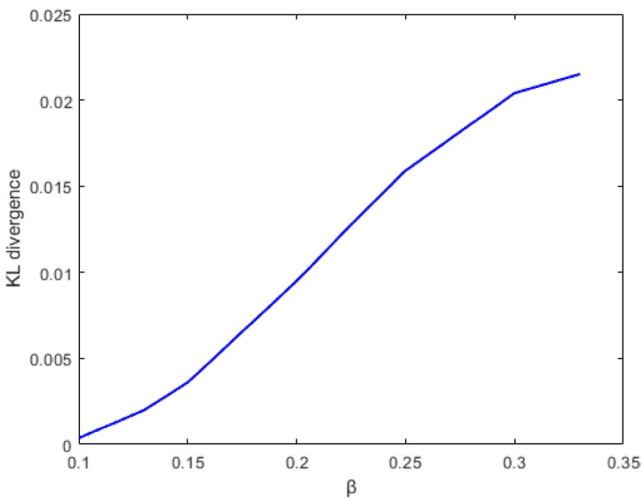

Figure 2. Kullback-Leibler divergence between Laguerre orthogonal polynomials and the K distribution for different β values.

If we let $1/\beta = \alpha/b$, and use Laguerre orthogonal polynomials, the PDF of the K distribution is approximated as (Tekel & Cohen, 2012).

$$f(h) = \beta e^{-\beta h}\left\{1 + \frac{1}{\alpha}\left\{\frac{1}{2}\beta^2 h^2 - 2\beta h + 1\right\}\cdots\right\} \tag{7}$$

Using the first and second terms in Equation 7 for convenience, there is

$$f(h) \approx \beta e^{-\beta h}\left\{1 + \frac{1}{\alpha}\left\{\frac{1}{2}\beta^2 h^2 - 2\beta h + 1\right\}\right\} \tag{8}$$

The KL divergence is used to evaluate the validity of the approximate distribution:

Generally, the KL divergence is used to measure the difference degree of between two distributions, and its mathematical expression is defined as follows (Singh et al., 2018).

$$D_{KL}(p_x(x)\|q_x(x)) = \int_0^\infty p_x(x)\ln(p_x(x)/q_x(x))dx \tag{9}$$

where $p_x(x)$ is the PDF of the exact distribution, and $q_x(x)$ is approximate PDF. In this paper, $p_x(x)$ and $q_x(x)$ correspond to the PDF in Equations 3 and 8, respectively. A smaller KL divergence, corresponds to more similar distributions. Conversely, the disparity between the two distributions is larger. Figure 2 shows that the KL divergence changes vary with β, while $\mu = 1$. Also, the KL divergence increases. Therefore, we can choose a smaller β to decrease the KL divergence to obtain a better approximate distribution of the K distribution.

3. GMSK Demodulation Over Strong Solar Wind Turbulence Channel

In deep space communication, the transmission data links are always subject to the low SNR, limited power and spectrum resources. As a class of constant envelope modulation techniques, GMSK modulation is recommended for deep space missions by CCSDS and the European Space Agency (ESA) (Blue Book, 2003; Mascarello et al., 2019). Therefore, we analyze the BER performance of GMSK demodulation over both strong solar wind turbulence and AWGN channels using MATLAB. Additionally, we use the BPSK modulation as a comparison. Viterbi algorithm is used for demodulation, because the optimal demodulator for a GMSK signal over AWGN implements the maximum likelihood sequence search algorithm. This phenomenon is known as the trellis demodulator when the sequence search is performed using the Viterbi algorithm (Lui & Tsai, 1999).

Figure 3 gives the BER performance of GMSK with $m = 1.2$, $m = 1.4$, $b = 1$ and $m = 1.2$, $m = 1.4$, $b = 2$, and the SNR ranges from −4 to 8 dB. In Figures 3a and 3b show that GMSK has better BER performance than BPSK over strong solar wind turbulence channel. Although the BER performance changes little in lower SNR, due to its high spectrum efficiency, which implies that GMSK as the modulation method for deep space exploration with low SNR is more suitable. Of course, the scintillation index has an obvious effect on the BER. Especially, when the SNR increases, although the scintillation index is small, the BER is also higher than that with no turbulence. When the scintillation index m decreases, which means that the turbulence effect becomes weaker, the BER is reduced.

Also, Figures 3a and 3b show that small b can achieve good BER performance.

For $m = 1.2$, Figure 4 compares the BER performance of GMSK and BPSK with different bandwidth-time products BT_b and b. The simulation results show that different BT_b values affect the system BER performance, and a better BER performance can be obtained when a small BT_b is taken for high a SNR. However, for a low SNR, the effect of BT_b on the BER performance is not obvious. Additionally, a large BT_b is chosen for low BER while high SNR. Hence, a small BT_b can be chosen for deep space communication with a low SNR due to its high spectrum efficiency.

4. Architecture and Mathematical Description of Neural Network Demodulator

Deep neural networks have been successfully applied to many important fields, such as computer vision (Krizhevsky et al., 2017), natural language processing (Cho et al., 2014), and speech signal processing (Weng

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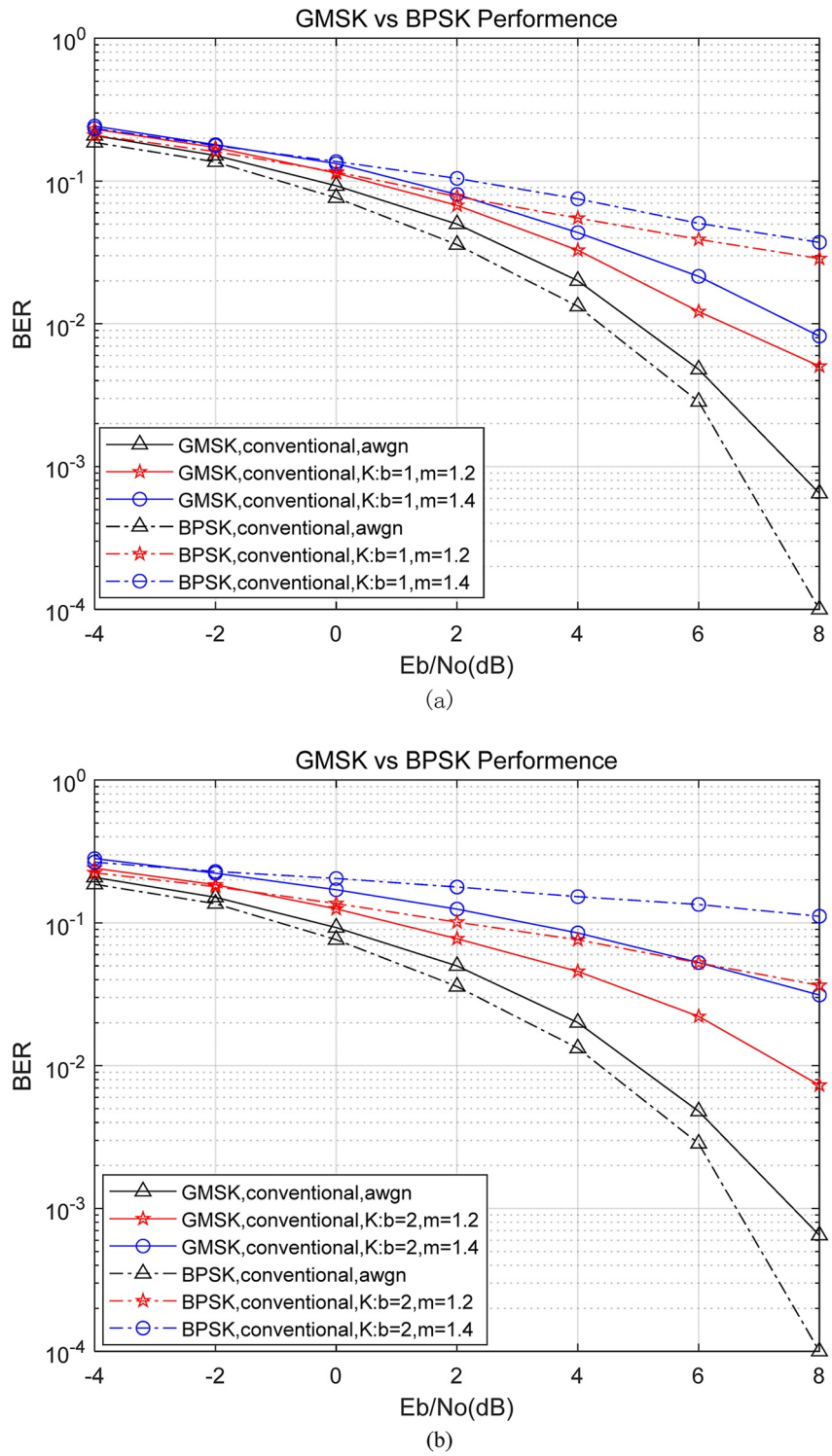

Figure 3. The bit error rate performance of Gaussian minimum frequency shift keying and binary phase shift keying demodulation for different b values.

et al., 2014) etc. Recently, there have been increasing many cases of applying deep learning to wireless communication systems, and they have shown good results. In this paper, we propose three neural network demodulators.

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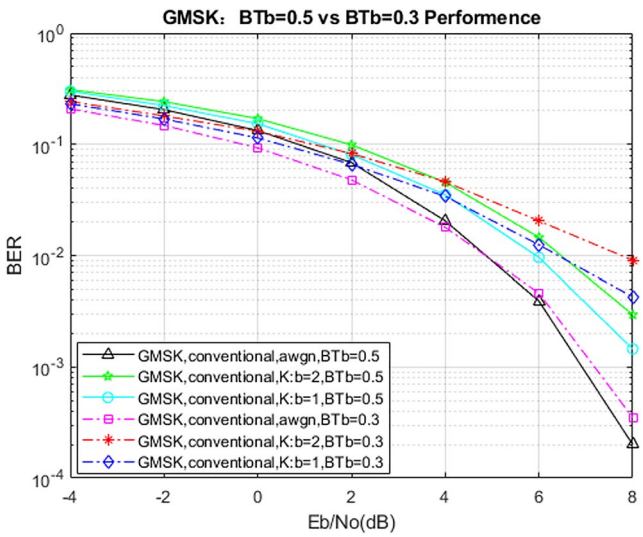

Figure 4. Bit error rate performance of Gaussian minimum frequency shift keying demodulation when $BT_b = 0.5$ and $BT_b = 0.3$.

4.1. Architecture of the 1D-CNN

CNN is a type of feed-forward neural network, that groups neurons in each layer into a convolution kernel and learns patterns from the input data (Indolia et al., 2018). Simultaneously, it can also reduce the dimensionality of feature data, which is very advantageous for larger dimensional data (Zhang et al., 2018). Therefore, we use a novel 1D-CNN to demodulate GMSK signals. The proposed 1D-CNN demodulator is depicted in Figure 5. The network consists of eight layers, including an input layer, two convolutional layers, a maximum pooling layer, three fully connected layers and an output layer. Among them, the convolutional layer extracts the local features of the GMSK signals through a 1×11 convolutional kernel, while filtering out another redundant feature. The pooling layer serves to downscale the features extracted by the convolutional layer. Finally, the pooled feature sequence is fed to the fully connected layers for classification, that is, the feature information is mapped to the interval [0, 1].

4.2. Architecture of the Bi-LSTM

LSTM (Adegboruwa et al., 2019) is an improved recurrent neural network (RNN), model that inherits the strong ability of RNN to process time series. Meanwhile, it also solves the problems of both gradient disappearance and gradient explosion when RNN processes long sequences. Different from the original LSTM, the Bi-LSTM can obtain sequence dynamic information from the past (backward) and future (forward) states (Wu, 2019), so it can better extract related sequence features. In this paper, the GMSK modulation signal is sampled from 100 sample points per symbol.

Utilizing the ability of Bi-LSTM to process sequence dynamic information, we propose a Bi-LSTM demodulator as shown in Figure 6, which consists of eight layers, including an input layer, two Bi-LSTM layers, three fully connected layers and an output layer. Two Bi-LSTM layers are used to learn patterns from the input data, then a max pooling layer is used to downscale the learning patterns, finally, they are sent to fully connected layers for classification.

4.3. Architecture of the Combining 1D-CNN With Bi-LSTM

Due to the 1D-CNN and LSTM have better extraction capability for both local features and sequence dynamic information of GMSK signals, a novel neural network that combining 1D-CNN and Bi-LSTM is designed for GMSK demodulation over strong solar wind turbulence channel. As shown in Figure 7, the proposed novel model combines 1D-CNN and Bi-LSTM to obtain a parallel architecture, which extracts the input data local features and temporal dynamic information through combining 1D-CNN and Bi-LSTM. Using the combining architecture, the features information of 1D-CNN is integrated with the temporal dynamic information of Bi-LSTM. And the

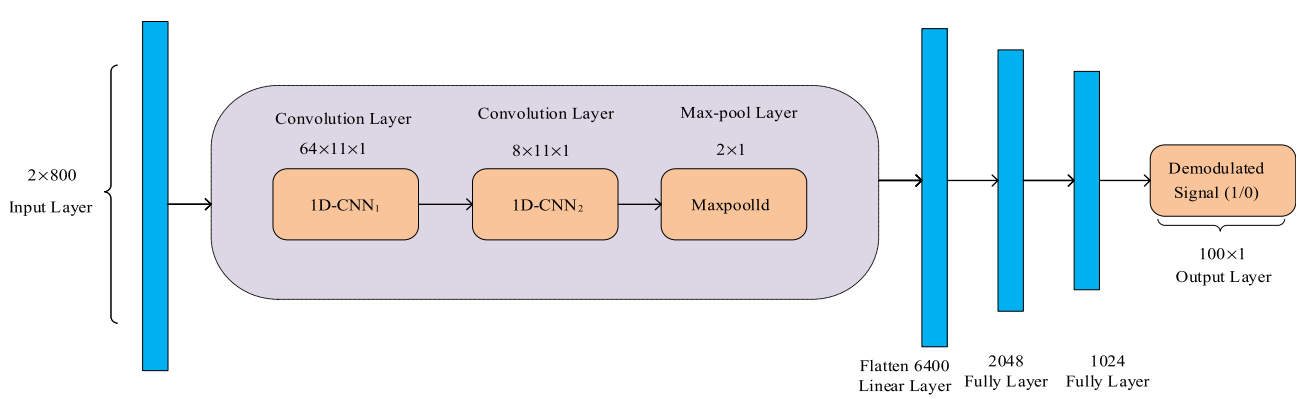

Figure 5. Architecture of one-dimensional convolutional neural network demodulator.

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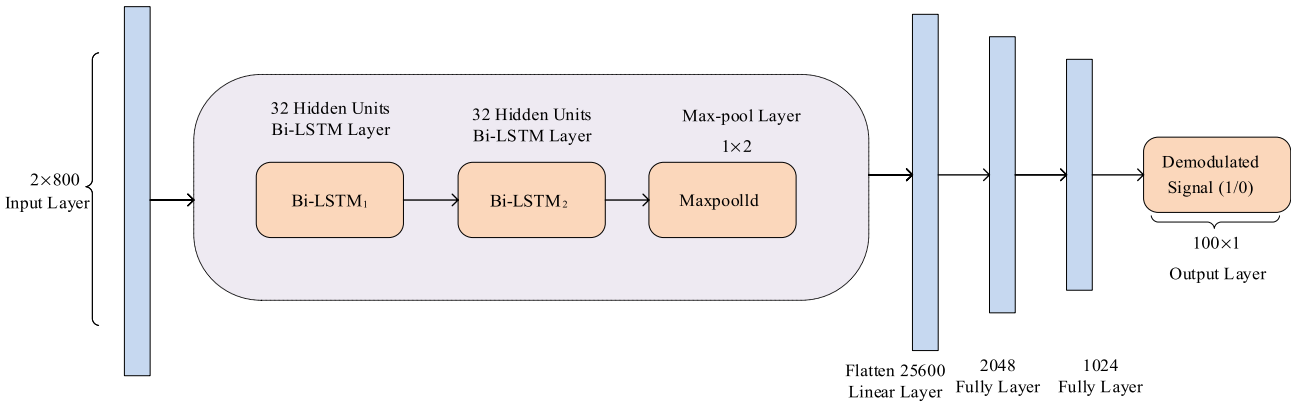

Figure 6. Architecture of bidirectional long short-term memory network demodulator.

parallel architecture is regarded as the input of the full connection layer. Finally, feature information is mapped to a 100-dimensional 0 and 1 binary classification through three full connection layers.

4.4. Mathematical Description

It is well known that the input of each hidden layer in the network architecture is a nonlinear function of the output of the previous layer. This kind of nonlinear function is often called the activation function, which adds nonlinear factors to the neural network model, and increases the generalization ability of the model. Here, the rectified linear unit (ReLU) and Sigmoid are adopted as activation functions for the hidden layer and output layer in the proposed model, respectively. which are expressed as follows.

$$f_R(x) = \max(0, x) \tag{10}$$

$$f_s(x) = \frac{1}{1 + e^{-x}} \tag{11}$$

where x is the output of the previous hidden layer.

Due to the parallel network architecture is only suitable for noncomplex numbers, we divide the input signals into real and imaginary parts. The received GMSK signals can be written as:

$$\begin{aligned} \mathbf{y}_j = \{&r(\mathrm{y}_j(0)),\ i(\mathrm{y}_j(0));\ r(\mathrm{y}_j(1)),\ i(\mathrm{y}_j(1)); \\ &\ldots\ r(\mathrm{y}_j(\mathrm{n})),\ i(\mathrm{y}_j(\mathrm{n}))\},\ j = 0, 1, \ldots \mathrm{M} - 1 \end{aligned} \tag{12}$$

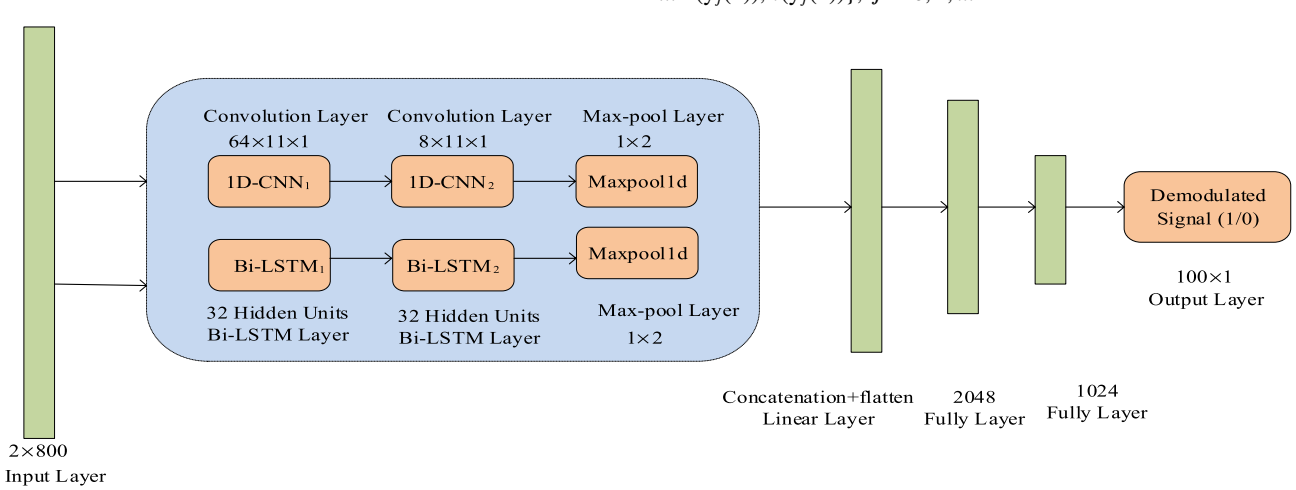

Figure 7. Architecture of combining one-dimensional convolutional neural network with bidirectional long short-term memory network demodulator.

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where $r(\cdot)$ and $i(\cdot)$ are the real part and the imaginary part, respectively. n is the length of the GMSK signal. Thus, the input of the hybrid network architecture can be expressed as:

$$\mathbf{I} = \{\mathbf{y_0}, \mathbf{y_1}, ..., \mathbf{y_{M-1}}\}^T \tag{13}$$

where M is the batch size of the input data. Then, the output feature vectors of the two feature extraction layers, the 1D-CNN and LSTM can be expressed as:

$$\mathbf{Z}_{\mathrm{CNN}} = f_R(\mathbf{W}_{\mathrm{CNN}}\mathbf{I} + \mathbf{b}_{\mathrm{CNN}}) \tag{14}$$

$$\mathbf{Z}_{\mathrm{LSTM}} = f_R(\mathbf{W}_{\mathrm{LSTM}}\mathbf{I} + \mathbf{b}_{\mathrm{LSTM}}) \tag{15}$$

where $\mathbf{W}_{\mathrm{CNN}}$ is the weight matrix of 1D-CNN layer, $\mathbf{W}_{\mathrm{LSTM}}$ is the weight matrix of the Bi-LSTM layer, $\mathbf{b}_{\mathrm{CNN}}$ and $\mathbf{b}_{\mathrm{LSTM}}$ are the bias vector of the 1D-CNN layer and Bi-LSTM layer, respectively. Then, two feature vector are connected and the fused feature vectors can be expressed as:

$$\mathbf{I}_{\mathrm{FC}} = \mathrm{flatten}(\mathrm{concat}(\mathbf{Z}_{\mathrm{CNN}}, \mathbf{Z}_{\mathrm{LSTM}})) \tag{16}$$

where, $\mathrm{flatten}(\cdot)$ is the feature vector expansion function, $\mathrm{concat}(\cdot)$ is the connection function of the feature vector, $\mathbf{I}_{\mathrm{FC}}$ is the input vector of fully connected layer.

The output of the network $\hat{\mathbf{Y}}$ is a cascade nonlinear transformation of input data vector $\mathbf{I}_{\mathrm{FC}}$ (Ye et al., 2018), and mathematically expressed as:

$$\hat{\mathbf{Y}} = f(\mathbf{I}_{\mathrm{FC}}, \mathbf{W}_{\mathrm{FC}}, \mathbf{b}_{\mathrm{FC}}) = f_S{}^{(\mathrm{L}-1)}\left(f_R{}^{(\mathrm{L}-2)}\left(\cdots f_R{}^{(1)}(\mathbf{W}_{\mathrm{FC}}\mathbf{I}_{\mathrm{FC}} + \mathbf{b}_{\mathrm{FC}})\right)\right) \tag{17}$$

where L is the number of layers, $\mathbf{W}_{\mathrm{FC}}$ is the weight matrix, and $\mathbf{b}_{\mathrm{FC}}$ is the bias vector of the fully connection layer. $\mathbf{W}_{\mathrm{FC}}$ and $\mathbf{b}_{\mathrm{FC}}$ are optimized in the backpropagation process. f is above mentioned functions, that is, f_R or f_S.

The neural network is trained to minimize the error between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ by continuously updating the values of $\mathbf{W}$ and $\mathbf{b}$. We select the mean square error function as the loss function in the training process, which is

$$J\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) = \frac{1}{N}\sum_{k=1}^{N}\left[\mathbf{Y}(k) - \hat{\mathbf{Y}}(k)\right]^2 \tag{18}$$

where $\mathbf{Y}$ is the label vector, and N is length of the sample.

Let $\theta = \{\mathbf{W}_{\mathrm{CNN}}, \mathbf{W}_{\mathrm{LSTM}}, \mathbf{W}_{\mathrm{FC}}, \mathbf{b}_{\mathrm{CNN}}, \mathbf{b}_{\mathrm{LSTM}}, \mathbf{b}_{\mathrm{FC}}\}$, then the objective function is

$$\theta = \arg\min_{\theta} J \tag{19}$$

Furthermore, we utilize the gradient descent algorithm to achieve the optimal θ, by

$$\theta(n+1) \leftarrow \theta(n) - \varepsilon\frac{\partial J\left(\mathbf{Y}, \hat{\mathbf{Y}}\right)}{\partial\theta} \tag{20}$$

where $\theta(n)$ is the weight and bias of the nth training, ε is the learning rate.

5. Neural Networks Demodulate Results and Analysis

In this section, we analyze the performance of GMSK demodulation using neural network over strong solar wind turbulence channel by using MATLAB and Python. Since there is no public data set for the strong solar wind turbulence channel, we use the public data set multi-task learning data set automatic modulation classification and DOA estimation from the IEEEDataPort to validate our proposed algorithm.

5.1. Generation Data Set and Setting Parameters

Generation of Training and Test Data Sets: the received signal of GMSK modulation over strong solar wind turbulence channel in Equation 2 is taken as the input layer data of the proposed neural network demodulator.

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Table 1
Neural Network Parameters Setting

Layer number	Layer type	Parameters	Layer number	Layer type	Parameters
1	1D-convolution	$64 \times 11 \times 1$	11	Bi-LSTM	Hidden units: 32
2	1D-batch normalization	64	12	Relu	/
3	Relu	/	13	Feature concatenation and flatten	28,800
4	1D-convolution	$8 \times 11 \times 1$	14	Fully connected layer	(28,800, 2048)
5	1D-batch normalization	8	15	Relu and dropout	Dropout rate: 0.08
6	Relu	/	16	Fully connected layer	(2,048, 1,024)
7	1D-MaxPooling	2×1	17	Relu and dropout	Dropout rate: 0.08
8	Relu	/	18	Fully connected layer	(1,024, 100)
9	Bi-LSTM	Hidden units: 32	19	Sigmoid	/
10	Tanh and dropout	Dropout rate: 0.2	20	output layer	100×1

The sampling number of each GMSK symbol is 8 and the number of transmission symbols is 100. Thus, the size of the input layer of the novel neural network in the proposed demodulation is 2×800. Also, the received signals are collected as test data, which is used to test our training model.

Simulation Parameters: scintillation index $m = 1.2$ and $m = 1.4$, the parameters of the solar wind turbulence channel $\alpha = 10$, $\alpha = 5$ and $b = 2$. The layer structure parameters of the hybrid network model are shown in Table 1.

The main parameters of the neural network training process are shown in Table 2.

5.2. Results and Analysis

The brief training process of the deep parallel network refers to Algorithm 1 as Table 3.

Using Equation 18, the relationship of the average loss value and training epochs is obtained, as shown in Figure 8a. Here, the optimizer is Adam (Khaire & Dhanalakshmi, 2020). In Figure 8a, the average loss values

Table 2
Related Parameters Setting

Parameters	Experimental parameter setting
Training data quantity	63,000
Validation data quantity	7,875
Test data quantity	4200
Input data size	2×800
Range of SNR	−4 to 8 dB
Loss function	Mean square error loss
Optimizer	Adam
Batch size	512
Training epoch	40
Dropout rate	0.08
Pytorch	GPU (1.7.0)
System environment	Ubuntu 18.04.5, Custom Graphics Workstation (UItraLAB GX630i RTX 2080Ti)

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Table 3
The Deep Parallel Network Training Algorithm

Algorithm 1 The Deep Parallel Network Training

1:**Input:** GMSK modulation data $\{\mathbf{I}\}$

2:**Output:** the category of information bit

3:**for** k = 1,2,...,N **do**

4: Forward propagation: input **I** into 1D-CNN layer and

5: Bi-LSTM layer, and continue to propagate forward after

6: feature fusion.

7: Calculate loss J

8: Back propagation: $\frac{\partial J(\mathbf{Y},\hat{\mathbf{Y}})}{\partial \boldsymbol{\theta}}$ be calculated and propagate

9: to the previous layer structure.

10: Update $\boldsymbol{\theta}$ with gradient descent algorithm.

11:**end for**

of the training and validation set are close to 0 after only 30 epochs of training. Also, Figure 8b shows that the validation and training accuracy are close to 1 as the number of training epochs increasing. Thus, the proposed model can quickly converge and exhibit no overfitting phenomenon.

For scintillation indices $m = 1.2$ and $m = 1.4$ respectively, Figure 9 compares the BER performance of the proposed neural network and conventional Viterbi demodulation at $BT_b = 0.3$. The simulation results show that the BER performance degrades when the scintillation index increases, and the proposed neural network demod-

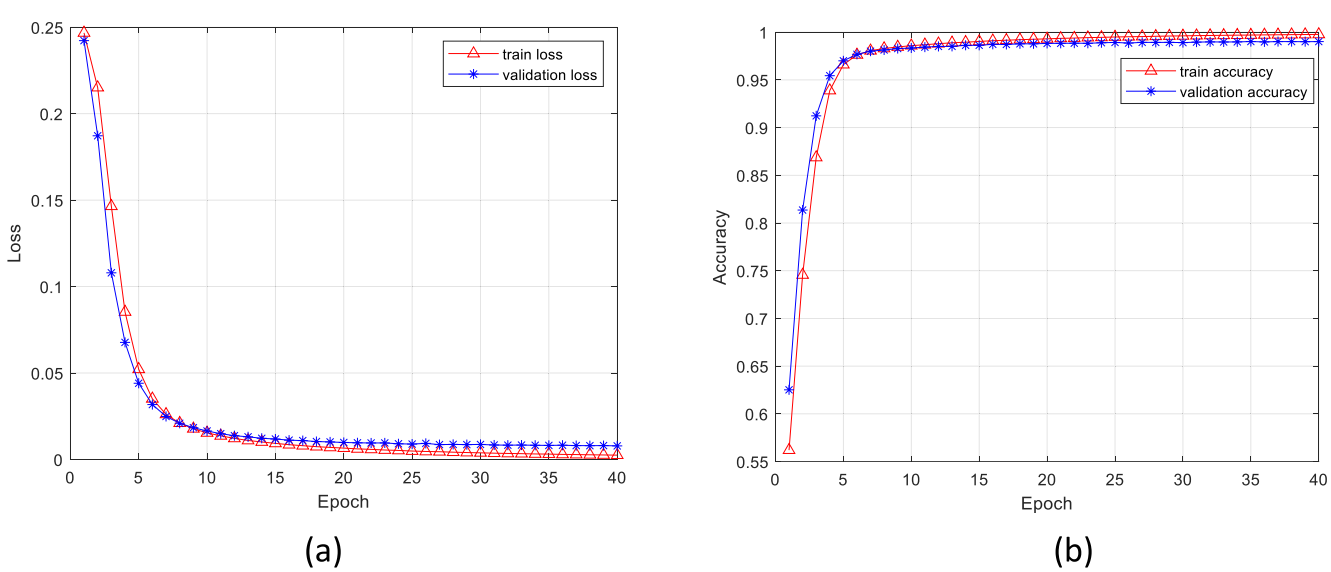

Figure 8. Average loss and accuracy in the process of training.

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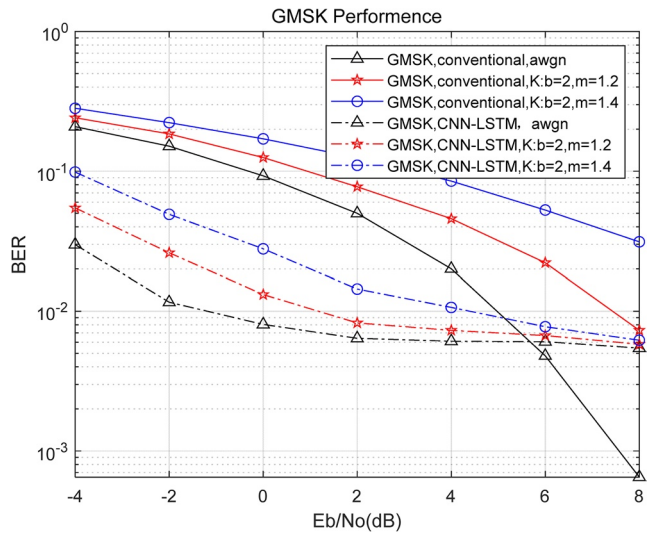

Figure 9. Bit error rate performance of deep learning and traditional demodulation at $BT_b = 0.3$.

ulation has an obviously better BER performance than the conventional Viterbi algorithm, especially when the SNR is lower. This result implies that the solar scintillation characteristics directly affect BER performance, and the proposed neural network demodulation is more suitable for deep space exploration of the solar system.

For scintillation indices $m = 1.2$ and $m = 1.4$, Figure 10 compares the BER performance of the proposed demodulation with $BT_b = 0.5$ and $BT_b = 0.3$, respectively. The results show that the demodulation performance with $BT_b = 0.5$ is better. However, the spectrum efficiency demodulation with $BT_b = 0.3$ is better. Hence, it is very important to select an appropriate BT_b value for a specific environment. Additionally, the effect of the solar scintillation characteristics on the BER performance is consistent with the results in Figure 9.

Finally, for scintillation indeices $m = 1.2$ and $m = 1.4$, Figure 11 compares the BER performance of three neural network demodulation, the demodulator with combining 1D-CNN and Bi-LSTM, only single 1-CNN and Bi-LSTM. The simulation results show that the combining model demodulation has the best BER performance at low SNR. since it is not enough sufficiently to extract the features of received signals only using 1D-CNN or LSTM in low SNR. The combining neural demodulator integrates the feature extraction capability of CNN and the processing capability of LSTM for time series, and it can obtain the best BER performance.

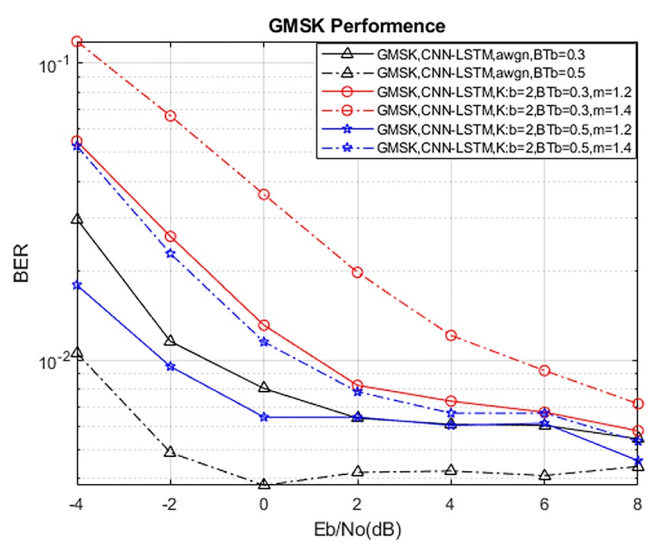

Figure 10. Bit error rate performance of deep learning demodulation at $BT_b = 0.3$ and $BT_b = 0.5$.

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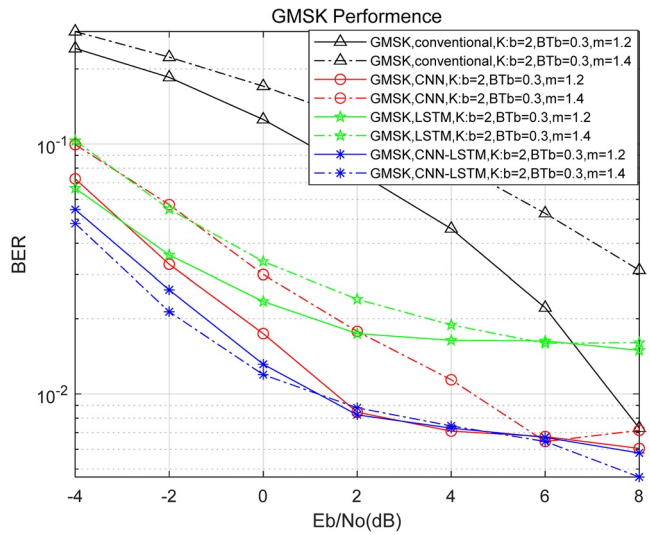

Figure 11. Bit error rate performance of different demodulations.

Also, other demodulators based deep neural network have better BER performance than the demodulator using Viterbi algorithm. From this point of view, demodulation based on deep neural networks will be a very promising direction in the application of complex strong solar wind turbulence channel. Meanwhile, the effect of the solar scintillation characteristics on the BER performance is keeping with the results both in Figures 9 and 10.

Also, compared with the traditional Viterbi algorithm, the proposed three neural network demodulators have higher flexibility, because the traditional coherent demodulator requires local oscillators to generate coherent carriers and bandpass filters to remove out-of-band noise. Therefore, using a deep neural network is an important path to simplify complex devices in deep space exploration.

5.3. Model Validation

To verify the practicality of the proposed neural network model, we obtain the public data set multi task learning data set automatic modulation classification and doa estimation from the IEEEDataPort. The data set is composed of various modulated signals generated by a wireless fading plus Gaussian noise channel at different SNR. The data set contains 450,120 signals, which covers 121 DOA classes, and 12 modulation classes at 31 SNR levels in the range [−10, 20] dB, where each signal has 1,024 samples. Among them, we extract the fading PSK modulated signals in the SNR in the range of [−4, 8] dB to validate our model. As shown in Figure 12, compared with the traditional Viterbi, CNN and LSTM algorithms, combining LSTM and CNN network architecture still obtains better BER performance. The practicality of the proposed network model is proven, which is applicable to the demodulation of fading signals under strong solar wind turbulence channel, and other wireless fading signals. The proposed neural network model has greater potential in the demodulation of wireless fading signals.

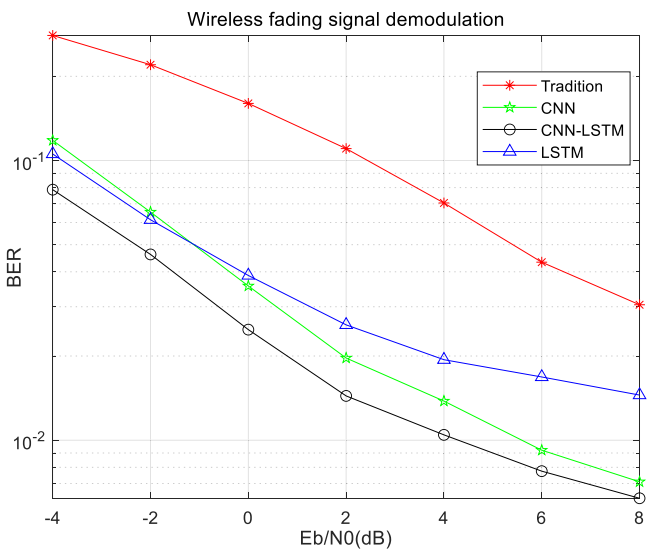

Figure 12. Bit error rate performance of model validation.

6. Conclusion

In this paper, we address to GMSK demodulation using ML over strong solar turbulence channel. We focus on the following: (a) By analyzing the scintillation characteristics of the telemetry signal caused by strong solar wind turbulence, we establish the K distribution channel model of the strong solar wind turbulence and give the approximate PDF of the K

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distribution of the established channel using the Laguerre orthogonal polynomial. (b) For the GMSK modulator by CCSDS, we present the BER performance of the GMSK demodulator using the Viterbi algorithm, a single 1D-CNN and an LSTM neural network. For different scintillation indices, the simulation results verify the effectiveness of demodulation using ML. Especially, combining the feature extraction capability of the 1D-CNN and the processing capability of LSTM for time series, we propose a novel GMSK demodulator, and the BER performance using the proposed novel GMSK demodulator improves even at low SNR. In addition, the public data set from IEEEDataPort is used to verify the proposed network model. The simulation results show that the proposed algorithm model is also suitable for demodulation of other wireless fading modulation signals, which can obtain better BER performance.

Conclusively, the proposed combining neural network demodulator for GMSK is more suitable for the deep space exploration of the solar system. The work in this paper helps to analyze and design a wireless communication system for deep space exploration of the solar system.

Data Availability Statement

All data sets in this paper are generated by our simulation using MATLAB. The simulation data sets are inspired by the analysis of the strong solar wind turbulence channel. The strong solar wind turbulence channel is modeled as a k-distribution model. GMSK fading signals that pass through the solar wind turbulence channel are collected as our data set and can be found at https://doi.org/10.5281/zenodo.5781913. This data set contains two files. The training data set is used to train the proposed neural network demodulator, and the test data is used to test trained neural network demodulator model. The public data set for verifying the network model can be found in IEEEDataPort. Public access to experiments: https://ieee-dataport.org/documents/multi-task-learning-dataset-automatic-modulation-classification-and-doa-estimation.

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